Sri Chaitanya IIT Academy., India.

A.P, TELANGANA, KARNATAKA, TAMILNADU, MAHARASHTRA, DELHI, RANCHI

A right Choice for the Real Aspirant

ICON Central Office – Madhapur – Hyderabad

Sec: **Sr.Super-60(In Coming)**	Jee-Main	Date: 04-09-2021
Time: 09.00Am to 12.00	**PTM-04**	Max.Marks:300

IMPORTANT INSTRUCTIONS:

1. Immediately fill in the Admission number on this page of the Test Booklet with Blue/Black Ball Point Pen only.

2. The candidates should not write their Admission Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.

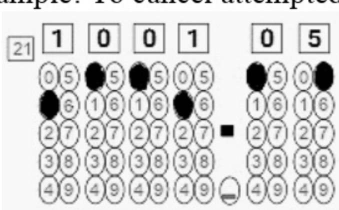
3. The test is of **3 hours** duration.

4. The Test Booklet consists of **90** questions. The maximum marks are **300**.

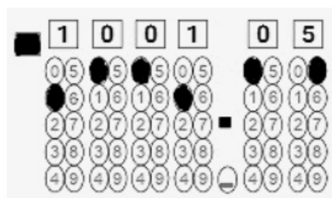
5. There are **three** parts in the question paper 1, 2, 3 consisting of **Physics**, **Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **two sections.**

(I) Section-I contains 20 **multiple choice** questions with **only one correct** option.
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

(II) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions only.
First 5 attempted questions will be considered if more than 5 questions attempted.
To cancel any attempted question bubble on the question number box.
For example: To cancel attempted question 21, bubble on 21 as shown below.



Question Answered for marking



Question Cancelled for marking

Marking scheme : +4 for correct answer and 0 in all other cases.

6. Use **Blue/Black Ball Point Pen only** for writing particulars/marking responses on the Answer Sheet. **Use of pencil is strictly prohibited**.

7. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall.

8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.

9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Hall. **However, the candidate are allowed to take away this Test Booklet with them.**

10. **Do not fold or make any stray marks on the Answer Sheet**

Name of the Candidate (in Capitals):____________________________________

Admission Number:

Candidate's Signature: ________________ **Invigilator's Signature:** ________________

04-09-2021_Sr.Super60(In Coming)_Jee-Main_PTM-04_Test Syllabus

PHYSICS : Electrostatics + Gravitation Coulomb's law; Electric field and potential; Electrical potential energy of a system of point charges and of electrical dipoles in a uniform electrostatic field; Law of gravitation; Gravitational potential and field; Acceleration due to gravity; Motion of planets and satellites in circular orbits; Escape velocity. (Excluding Gauss law, conductors)

CHEMISTRY : **Alcohol ; phenol & Ether Alcohols** : esterification, dehydration and oxidation, reaction with sodium, phosphorus halides, /concentrated HCl, **Phenols** : Acidity, electrophilic substitution reactions(halogeneation, nitration and sulphonation) ; Reimer-Tiemen reaction, Kolbe reaction. **Ethers** : Preparation, Properties & Reactions

MATHEMATICS : Ellipse & Hyperbola

PHYSICS **Max Marks: 100**

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

1. Two equally charged identical metal spheres A and B repel each other with a force of $2\times10^{-5}N$. If another identical uncharged sphere C is touched to B and then placed at the mid point between A and B then the net force on C is

1) $10^{-5}N$ along AB 2) $2\times10^{-5}N$ along AB
3) $2\times10^{-5}N$ along CA 4) $10^{-5}N$ along CA

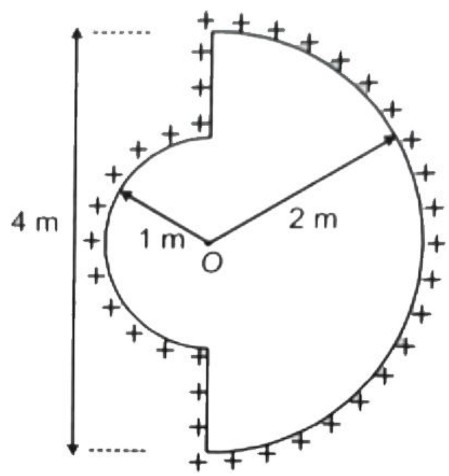
2. Two equally charged spheres of same mass are suspended from the same point by threads of same length. When the entire system is immersed in some liquid (d = 10 gm/cc; k=6), divergence angle between the threads does not change. Find the density of the sphere material

1) 16 gm/cc 2) 10gm/cc 3) 12gm/cc 4) 1gm/cc

3. If mass M is split into two parts m and (M-m) which are then separated by a distance, the ratio of $\frac{m}{M}$ that maximizes the gravitational force between the two parts is

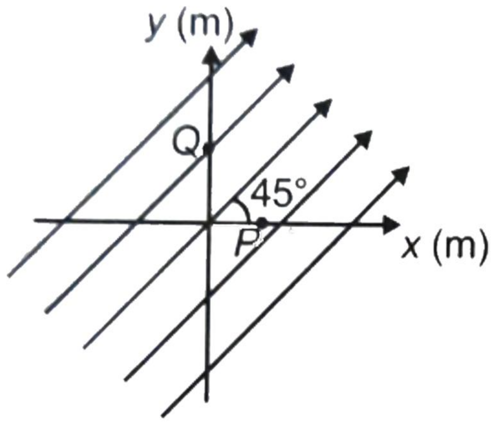
1) 1: 2 2) 1: 1 3) 2:1 4) 1:4

4. If $\lambda=1\mu$ C/m, then electric field intensity at O is



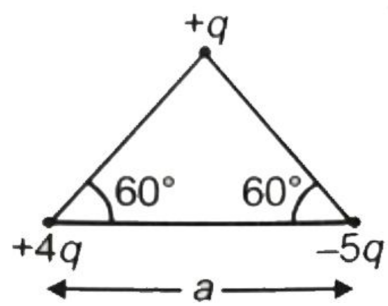
1) 9 N/C 2) 900 N/C 3) 9000 N/C 4) $9\times10^{9}N/C$

5. A uniform electric field of 200V/m is directed at 45° with x-axis as shown in figure. The co-ordinates of point P and point Q are (1,0) and (0,2). Find the potential difference, V_P-V_Q (in volts).



1) $\frac{200}{\sqrt{2}}$ 2) $200\sqrt{2}$ 3) $\frac{-200}{\sqrt{2}}$ 4) $-200\sqrt{2}$

6. Find the magnitude of dipole moment of the following system



1) $\sqrt{21}aq$ 2) zero 3) 4aq 4) 10 aq

7. Two short dipoles, each of dipole moment p, are placed at origin. The dipole moment of one dipole is along x-axis, while that of other is along y-axis. The electric field at a point (a,0) is given by

1) $\left(\frac{1}{4\pi \in_0}\right)\frac{2p}{a^3}$ 2) $\left(\frac{1}{4\pi \in_0}\right)\frac{p}{a^3}$ 3) $\left(\frac{1}{4\pi \in_0}\right)\frac{\sqrt{5}p}{a^3}$ 4) zero

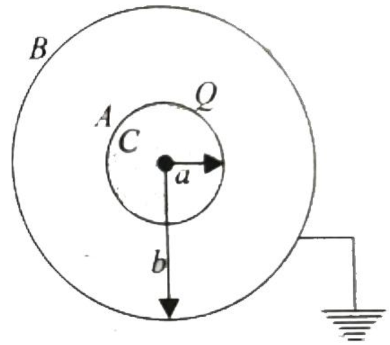
8. Two charges q and -4q are kept on a straight line. At how many points on this line, the electric potential is zero

1) 1 2) 2 3) 3 4) infinite

9. If the electric potential on the axis of an electric dipole at a distance 'r' from it is V, then the potential at a point on its equatorial line at the same distance away from it will be

1) 2V 2) $\frac{V}{2}$ 3) 0 4) –V

10. A conducting sphere A of radius a, with charge Q, is placed concentrically inside a conducting shell B of radius b. B is earthed. C is the common center of A and B. Study the following statements



i) The potential at a distance r from C, where $a \le r \le b, is \frac{1}{4\pi\varepsilon_0}\left(\frac{Q}{r}\right)$

ii) The potential difference between A and B is $\frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{a}-\frac{1}{b}\right)$

iii) The potential at a distance r from C, where $a \le r \le b, is \frac{1}{4\pi\varepsilon_0}Q\left(\frac{1}{r}-\frac{1}{b}\right)$

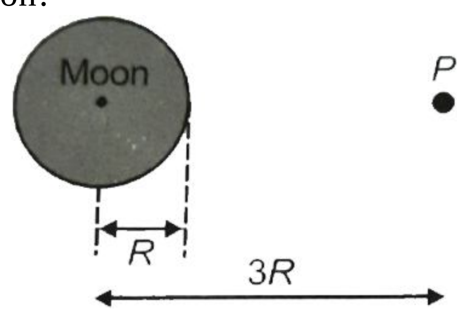
Which of the following statements are correct

1) Only i and ii 2) Only ii and iii 3) Only i and iii 4) All

11. If the change in the value of 'g' at a height 'h' above the surface of the earth is same as at a depth x below it, then (x and h being much smaller than the radius of the earth)

1) x = h 2) x = 2h 3) $x = \frac{h}{2}$ 4) $x = h^2$

12. A stationary object is released from a point P at a distance 3R from the centre of the moon which has radius R and mass M. Which of the following gives the speed of the object on hitting the moon?



1) $\left(\frac{2GM}{3R}\right)^{1/2}$ 2) $\left(\frac{4GM}{3R}\right)^{1/2}$ 3) $\left(\frac{GM}{3R}\right)^{1/2}$ 4) $\left(\frac{GM}{R}\right)^{1/2}$

13. Distance between centres of two stars is 10a. The masses of these stars are M and 16M and their radii are a and 2a, respectively. A body is fired straight from the surface of the larger star towards the smaller star. What should be its minimum initial speed to reach the surface of the smaller star?

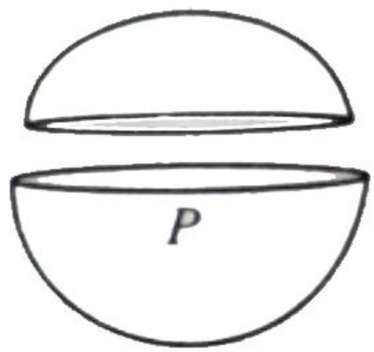
1) $\frac{3}{2}\sqrt{\frac{5GM}{a}}$ 2) $\sqrt{\frac{2GM}{a}}$ 3) $\frac{1}{2}\sqrt{\frac{5GM}{a}}$ 4) $\frac{3}{2}\sqrt{\frac{5GM}{2a}}$

14. A skylab of mass m kg is first launched from the surface of the earth in a circular orbit of radius 2r (from the centre of the earth) and then it is shifted from this circular orbit to another circular orbit of radius 3R. The minimum energy required to place the lab in the first orbit and to shift the lab from first orbit to the second orbit are

1) $\frac{3}{4}mgR, \frac{mgR}{6}$ 2) $\frac{3}{4}mgR, \frac{mgR}{12}$

3) mgR, mgR 4) 2mgR, mgR

15. A spherical shell is cut into two pieces along a chord as shown in the figure. P is a point on the plane of the chord. The gravitational field at P due to the upper part is I_1 and that due to the lower part is I_2. What is the relation between them?

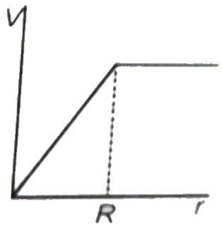


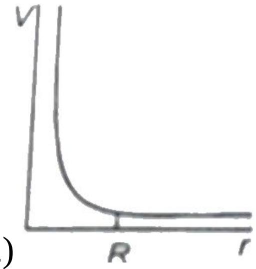
1) $I_1 > I_2$ 2) $I_1 < I_2$ 3) $I_1 = I_2$ 4) no definite relation

16. A satellite is moving with a constant speed v in a circular orbit about the earth. An object of mass m is ejected from the satellite such that it just escapes from the gravitational pull of earth. At the time of its ejection, K.E. of the object is

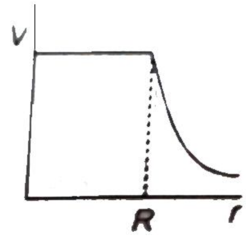
1) $\frac{1}{2}mv^2$ 2) mv^2 3) $\frac{3}{2}mv^2$ 4) $2mv^2$

17. A spherically symmetric gravitational system of particles has a mass density $\rho = \begin{cases} \rho_0 & for\ r \le R \\ 0 & for\ r > R \end{cases}$ Where ρ_0 is a constant. A test mass can undergo circular motion under the influence of the gravitational field of particles. Its speed V as a function distance $r(0 < r < \infty)$ from the centre of the system is represented by

1) 

2) 

3) 

4) 

18. In a planetary motion the areal velocity of a position vector of a planet depends on angular velocity ω and distance of the planet from the sun (r). If so the correct relation for areal velocity

1) $\frac{dA}{dt}\alpha\omega r$ 2) $\frac{dA}{dt}\alpha\omega^2 r$ 3) $\frac{dA}{dt}\alpha\omega r^2$ 4) $\frac{dA}{dt}\alpha\sqrt{\omega r}$

19. If two charges +q and +4q are separated by a distance 'd' and a point charge Q is placed on the line joining the above two charges and in between them such that all charges are in equilibrium. Then the charge Q and it's position are

1) $-\frac{4q}{9}$ at a distance $\frac{d}{3}$ from 4q 2) $\frac{-2Q}{3}$ at a distance $\frac{d}{3}$ form q

3) $-\frac{4q}{9}$ at a distance $\frac{d}{3}$ from q 4) $\frac{-2Q}{3}$ at a distance $\frac{d}{3}$ from 4q

20. Positive and negative point charges of equal magnitude are kept at $\left(0,0,\frac{a}{2}\right) and \left(0,0,\frac{-a}{2}\right)$, respectively. The work done by the electric field when another positive point charge is moved from $(-a,0,0)\ to\ (0,a,0)$ is

1) positive
2) Negative
3) Zero
4) Depends on the path connecting the initial and final positions

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

21. The time period of earth's rotation should be $x\pi\sqrt{\frac{R_e}{g}}$ if the weight of a body at equator become 3/4 times of its actual weight. Find the value of x.

22. A tunnel is made across the earth at distance $\frac{R}{2}$ from the centre of earth. A particle is dropped (from rest) in the tunnel at one end. If v_1 is the speed of particle when passing through the centre of tunnel and v_2 is the orbital speed of an earth satellite(very close to earth's surface). If $\frac{v_1}{v_2}=\frac{\sqrt{x+1}}{x}$. Find the value of x.

23. If w_1 joule work is done against gravitational attraction, to carry 10 kg mass from earth's surface to infinity. Then the magnitude of workdone by gravitational attraction in bringing 20kg mass from infinity to the centre, without any acceleration is w_2. Now if $w_2 = n\times w_1$. Find the value of n.

24. Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the earth. If the escape speed on the surface of the earth is taken to be 11 kms^{-1}, the escape speed on the surface of the planet is kms^{-1} will be

25. A particle of mass m is subjected to an attractive central force of magnitude k/r^2, k being a constant. If at the instant when the particle is at an extreme position in its closed orbit, at a distance *a* from the centre of force, its speed is (k/2 ma), if the distance of other extreme position is *b*. Find *a/b*

26. If the radius of the earth were to shrink by 2% its mass remaining the same, by how much percentage would the acceleration due to gravity on the earth's surface would increase?

27. There is an electric field E in x direction. If the work done in moving a charge of 2C through a distance of 2m along a line making an angle of 60^{o} with X axis is 3J, then the value of E is P N/c. What is the value of P ?

28. If gravitational forces between a planet and a satellite is proportional to $R^{-5/2}$ if R is the orbit radius. Then the squares of period of revolution of satellite is proportional to R^{n}. Find n.s

29. Consider an electric dipole located in uniform electric field in stable equilibrium position. It is now slowly rotated to the position of unstable equilibrium. Work alone by the external agent in the process is numerically how many times the maximum electric torque experienced by the dipole during the process?

30. The linear charge density on a dielectric ring of radius R vanes with $\theta \; as \; \lambda = \lambda_0 \cos\theta / 2$, where λ_0 is constant. Find the potential at the center O of the ring [in volt].

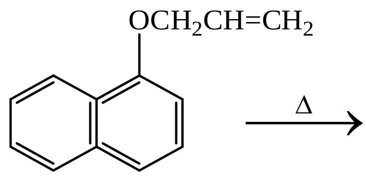
CHEMISTRY **Max Marks: 100**

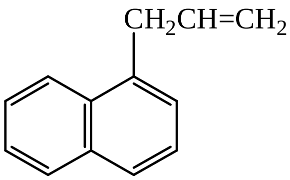
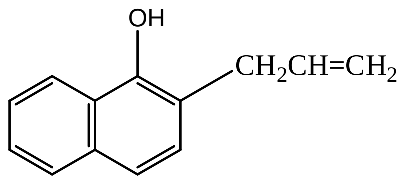
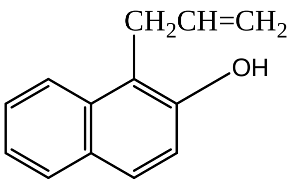
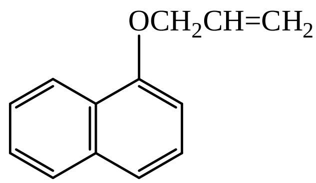
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

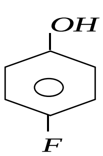
Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

31. The product formed in the given reaction is z



1) 
2) 
3) 
4) 

32. The correct order of acidic strength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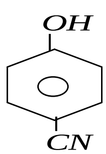
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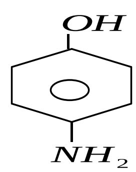
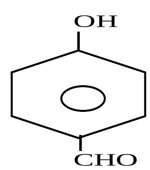

(M)

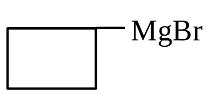

(N)

1) N>L>K>M
2) L>N>K>M
3) L>K>N>M
4) N>K>L>M

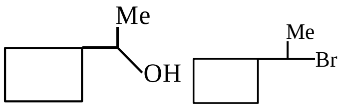
33. An organic compound C_7H_8O is soluble in $NaOH$. When treated with Br_2/H_2O- it gives a tribromo derivative .It is

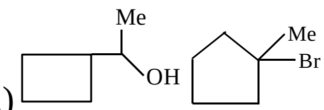
1) Benzyl alcohol
2) o-cresol
3) m-cresol
4) Anisole

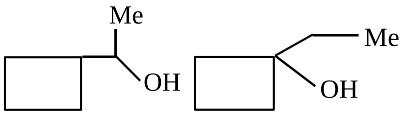
34.

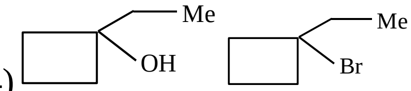


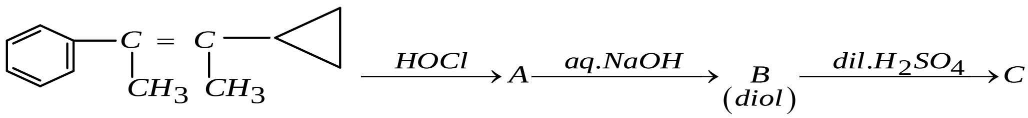
$\xrightarrow[2)\,H_3O^+]{1)\,CH_3-CHO}(A)\xrightarrow{HBr}(B)$. The compounds (A) and (B) are

1) 

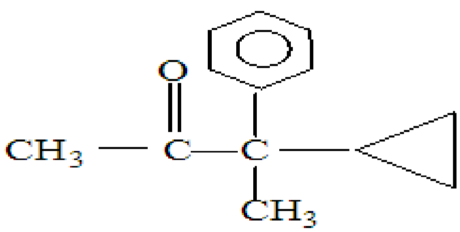
2) 

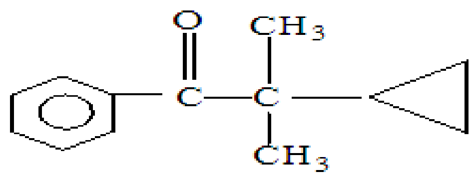
3) 

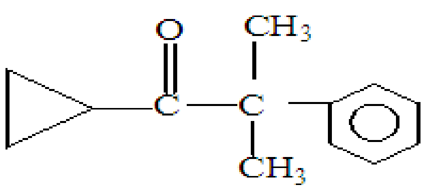
4) 

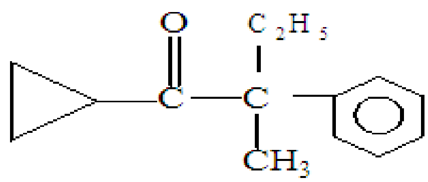
35. 

The product C is:

1) 

2) 

3) 

4) 

36. The decreasing order of reactivity of the following towards williamson synthesis is

P) $CH_3-\underset{CH_3}{\overset{CH_3}{C}}-\underset{Br}{CH_2}$

Q) $CH_3-CH_2-\underset{Br}{CH_2}$

R) $CH_3-CH_2-\underset{Cl}{CH_2}$

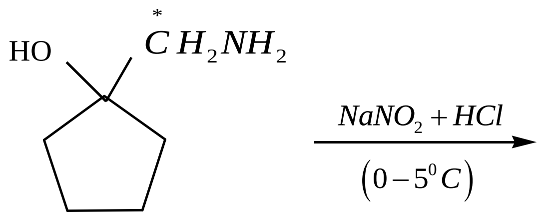
S) $CH_2=CH-\underset{Cl}{CH_2}$

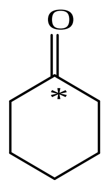
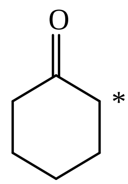
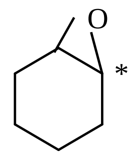
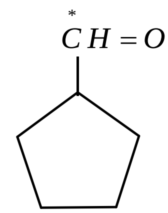
1) $P>Q>R>S$
2) $Q>R>S>P$
3) $P>R>Q>S$
4) $S>Q>R>P$

37. In the Libermann's nitroso reaction, sequential changes in the colour of phenol occurs as:

1) Brown or red $\rightarrow$ green $\rightarrow$ deep blue
2) Red $\rightarrow$ deep blue $\rightarrow$ green
3) Red $\rightarrow$green $\rightarrow$ colour less
4) White $\rightarrow$ red $\rightarrow$ green

38. The product P (major) of the following reaction is



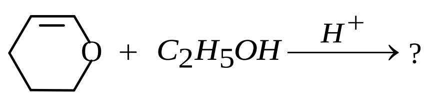
1)  2)  3)  4) 

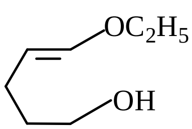
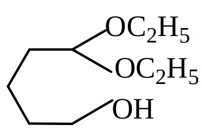
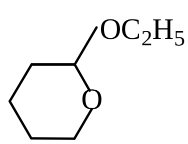
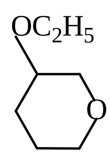
39. How many moles of HCOOH is formed , due to the oxidation with HIO_4 of,

$$R-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_2OH$$

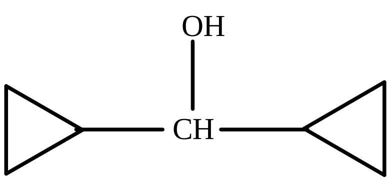
1) 1 2) 2 3) 3 4) 4

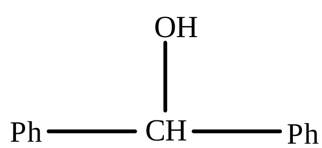
40. Which of the following is the product from ethanol addition to dihydropyran?

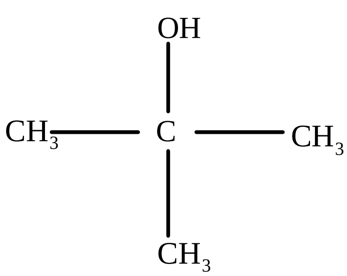


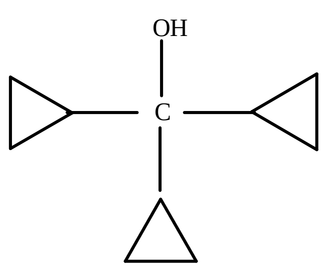
1)  2)  3)  4) 

41. Which one of the following will be most reactive for SN^1 reaction?

1) 

2) 

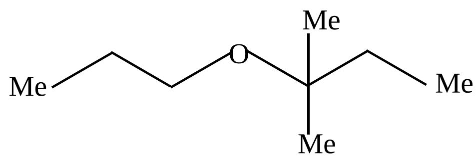
3) 

4) 

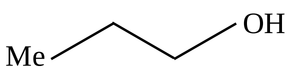
42. How many isomers of $C_4H_{10}O$ reacts with CH_3MgBr to evolve CH_4 gas? Excluding stereoisomer.

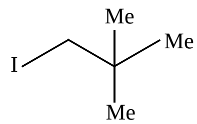
1) 3 2) 4 3) 5 4) 6

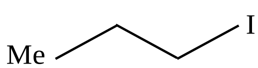
43.

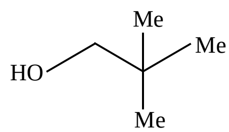


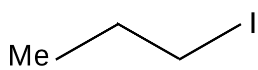
$\xrightarrow[\text{HI}]{\text{excess of}} (A)+(B)$ The products (A) and (B) are

1) 

and 

2) 

and 

3) 

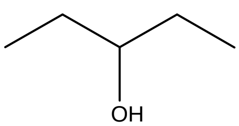
and $CH_3-CH_2-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-I$

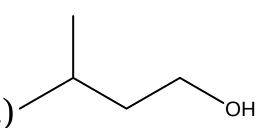
4) All

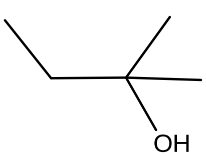
44 $CH_3CH_2OH \xrightarrow{?} CH_3CH_2Br$

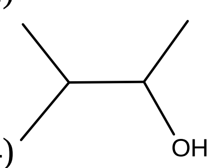
1) KBr 2) Br_2/hv 3) HBr 4) HOBr

45 Neo pentyl chloride on reaction with aqueous KOH preferably gives ______

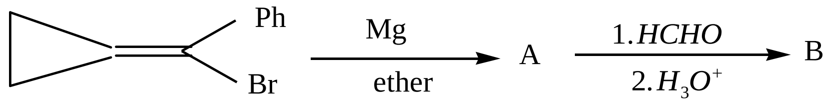
1) 

2) 

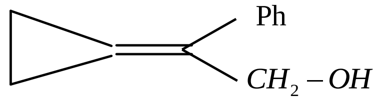
3) 

4) 

46.



: Product (B) is:

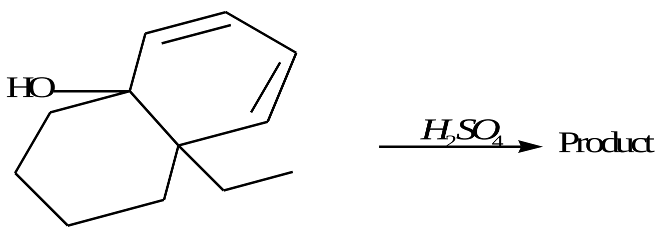
1) 

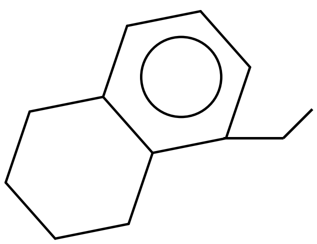
2) $Ph-C\equiv C-CH_2-CH_2-CH_2-OH$

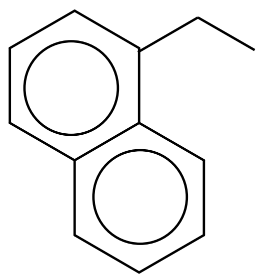
3) $Ph-C\equiv C-CH_2-CH_2-OH$

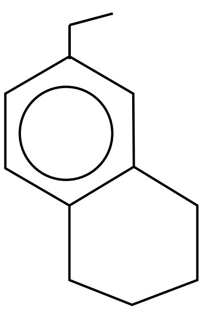
4) $Ph-CH_2-C\equiv C-CH_2-CH_2-OH$

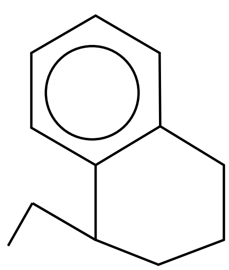
47. Identify the major product



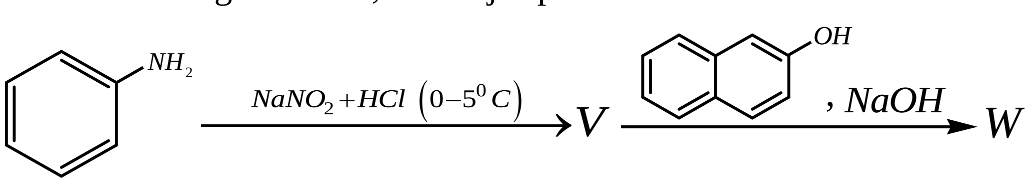
1) 

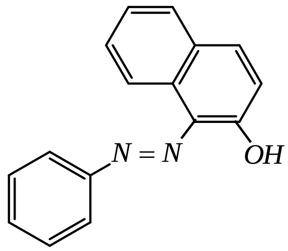
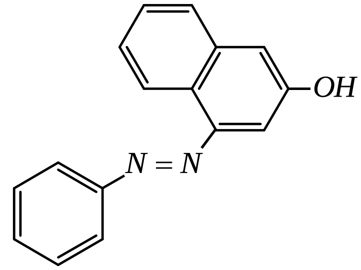
2) 

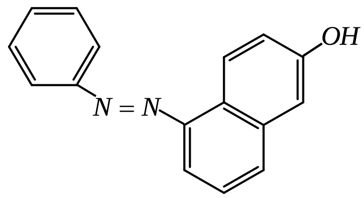
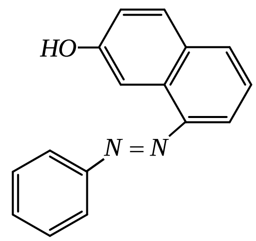
3) 

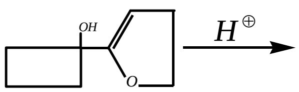
4) 

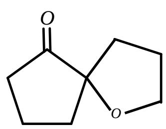
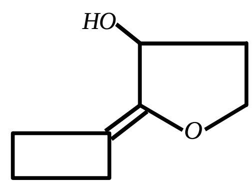
48. In the following reactions, the major product W is

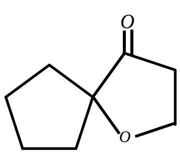
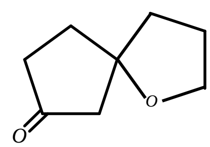


1)  2) 

3)  4) 

49.  (X) , The product (X) is:

1)  2) 

3)  4) 

50. 0.315 gm of organic compound having methoxy groups has molecular weight 235. This compound when treated with HI followed by $AgNO_3$ gave 0.468 of AgI. The percentage of methoxy group in the compound is

1) 19.6 2) 29.6 3) 1.96 4) 40

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

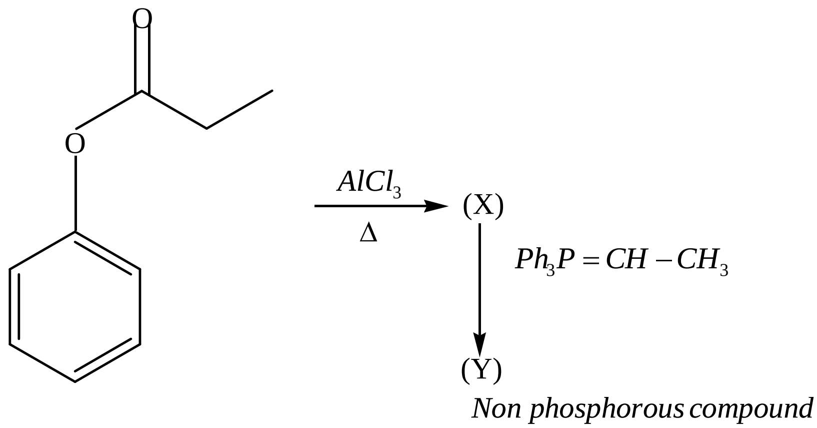
Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

51. How many cyclic isomers including stereo isomers exist, with the formula C_3H_6O ?

52. How many isomeric benzene derivatives exist with the formula C_7H_8O ?

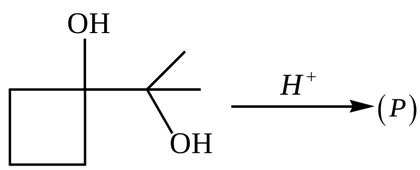
53. 0.92 g of a compound with molecular weight 92 releases 0.672 lit of methane at STP when treated with CH_3MgI. The number of OH groups it has is

54.

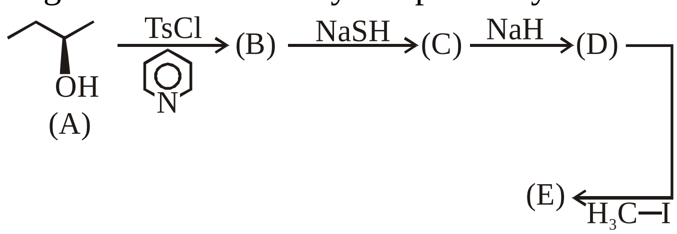


Degree of unsaturation of Y is "n" identify the value of "n"

55.



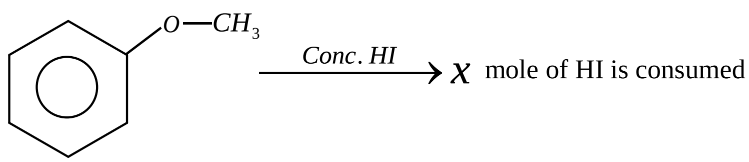
Number of methyl and hydroxyl substituents over the ring of P are "x" and "y" respectively. Therefore "x+y" is.

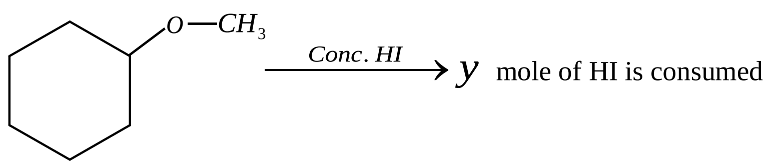


56.

The sum of C, O, and S atoms present compound (E) is "x". The value of "x" is

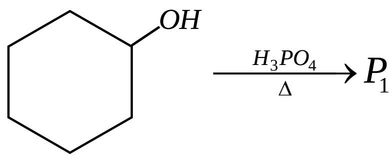
57.

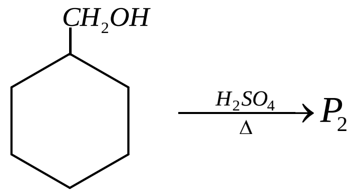


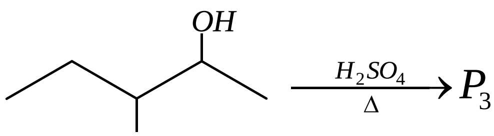


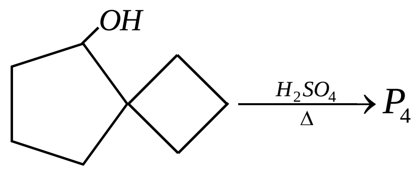
Sum of $(x+y=?)$:

58. In the following reactions, sum of α–hydrogen in major products $(P_1+P_2+P_3+P_4=?)$

a) 

b) 

c) 

d) 

59. I_2 produced when ozone reacts with moist KI is used to convert C_2H_5OH to CI_3CHO. Number of moles of ozone required to convert 1 mole of C_2H_5OH into CI_3CHO is

60. How many moles of $LiAlH_4$ is required, in order to convert 12 moles of Ethanoic acid to Ethyl alcohol

MATHS **Max Marks: 100**

(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and –1 in all other cases.

61. If a tangent of slope 2 of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is normal to the circle $x^2+y^2+4x+1=0$,then the maximum value of ab is

1) 1 2) 2 3) 4 4) 8

62. If the normal at an end of a latusrectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ passes through one extremity of the minor axis, then $e^2=$___

1) $2\sin 36^0$ 2) $2\cos 18^0$ 3) $2\sin 54^0$ 4) $2\cos 72^0$

63. If $\frac{x^2}{f(4a)}+\frac{y^2}{f(a^2-5)}=1$ represents an ellipse with major axis as y-axis and f is a decreasing function, then

1) $a\in(-\infty,1)$ 2) $a\in(5,\infty)$ 3) $a\in(-1,5)$ 4) $a\in(1,4)$

64. If $a=\left[t^2-3t+4\right]$ and $b=[3+5t]$, where $[.]$ denotes the greatest integer function , then the latusrectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ at $t=\frac{3}{2}$ is

1) 20 2) $\frac{1}{5}$ 3) 10 4) $\frac{1}{10}$

65. A straight line touches the rectangular hyperbola $9x^2-9y^2=8$ and the parabola $y^2=32x$, then the equation of the line is

1) $9x-3y-8=0$ 2) $9x-3y+8=0$

3) $9x+2y+8=0$ 4) $9x+3y-8=0$

66. The equation of a hyperbola, conjugate to the hyperbola $x^2+3xy+2y^2+2x+3y=0$ is

1) $x^2+3xy+2y^2+2x+3y+1=0$ 2) $x^2+3xy+2y^2+2x+3y+2=0$

3) $x^2+3xy+2y^2+2x+3y+3=0$ 4) $x^2+3xy+2y^2+2x+3y+4=0$

67. The condition that a straight line with slope m will be normal to parabola $y^2=4ax$ as well as a tangent to rectangular hyperbola $x^2-y^2=a^2$ is

1) $m^6-4m^2+2m-1=0$ 2) $m^4-3m^3+2m+1=0$

3) $m^6+2m^4+m^2-1=0$ 4) $m^6+4m^4+3m^2+1=0$

68. If the eccentricity of the hyperbola $x^2-y^2\sec^2\alpha=5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\alpha+y^2=25$, then the value of α is____

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{4}$

69. The locus of the mid points of the chord of the circle $x^2+y^2=25$, which is tangent to the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$ is

1) $(x^2+y^2)^2-16x^2+9y^2=0$
2) $(x^2+y^2)^2-9x^2+16y^2=0$
3) $(x^2+y^2)^2-9x^2+144y^2=0$
4) $(x^2+y^2)^2-9x^2-16y^2=0$

70. Sum of the eccentric angles of a point on the ellipse $\frac{x^2}{6}+\frac{y^2}{2}=1$, where distance from the centre of the ellipse is $\sqrt{5}$

1) 4π
2) 2π
3) π
4)0

71. The point of intersection of the curve whose parametric equations are $x=k^2+1, y=2k$ and $x=2t, y=\frac{2}{t}$ is given by

1) $(1,3)$
2) $(-2,4)$
3) $(2,2)$
4) $(3,2\sqrt{2})$

72. The set of values of a for which $(13x-1)^2+(13y-2)^2=a(5x+12y-1)^2$ represents an ellipse, if ___

1) $1<a\le 2$
2) $0<a<1$
3) $2<a\le 3$
4) $0\le a<1$

73. The eccentricity of an ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the axis of x and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the axis of y and whose axis lie along the axis of co-ordinates is ____

1) $\frac{3\sqrt{2}}{7}$
2) $\frac{2\sqrt{3}}{7}$
3) $\frac{\sqrt{3}}{7}$
4) none of these

74. A tangent drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ at $P\left(\pi/6\right)$ from a triangle of area $3a^2$ square units with coordinate axis. The eccentricity of the conjugate hyperbola is ___

1) $\sqrt{17}$
2) $\frac{\sqrt{17}}{4}$
3) $\frac{\sqrt{17}}{2}$
4) $\frac{8}{\sqrt{17}}$

75. If an ellipse has foci $(4,2),(2,2)$ and it passes through the point $P(2,4)$, then eccentricity of an ellipse is____

1) $\tan\frac{\pi}{10}$
2) $\tan\frac{\pi}{12}$
3) $\tan\frac{\pi}{6}$
4) $\tan\frac{\pi}{8}$

76. If $\alpha+\beta=3\pi$, then the chord joining the points α and β for the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ passes through

1) focus
2) centre
3) one end of the transverse axis
4) one end of the conjugate axis

77. Latusrectum of the conic satisfying the differential equation $xdy+ydx=0$ and passing through the point $(2,8)$ is___

1) $4\sqrt{2}$ 2) 8 3) $8\sqrt{2}$ 4) 16

78. The locus of a point $P(\alpha,\beta)$ moving under the condition that the line $y=\alpha x+\beta$ is a tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is

1) ellipse 2) circle 3) parabola 4) hyperbola

79. For the hyperbola $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$, which of the following remains constant when α varies

1) abscissa of vertices 2) abscissa of foci

3) eccentricity 4) directrix

80. The maximum area of triangle formed by the tangent to an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and coordinate axis is __

1) $\frac{a^2+b^2}{2}$ 2) ab 3) $\frac{(a+b)^2}{2}$ 4) $\frac{a^2+ab+b^2}{3}$

(NUMERICAL VALUE TYPE)

This section contains 10 questions. Each question is numerical value type. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to second decimal place. (e.g. 6.25, 7.00, 0.33, 30, 30.27, 127.30).

Attempt any five questions out of 10.

Marking scheme: +4 for correct answer, 0 if not attempted and 0 in all other cases.

81. If a tangent of slope 3 of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{1}=1$ passes through the point $(-2,0)$ then the value of $\left[a^2\right]$ is ____ (Where [.] denotes G.I.F)

82. If the normal to the ellipse $\frac{x^2}{25}+\frac{y^2}{1}=1$ is at a distance P from the origin then the maximum value of P is____

83. If P and Q are points with eccentric angles θ and $\left(\theta+\frac{\pi}{6}\right)$ on the ellipse $\frac{x^2}{16}+\frac{y^2}{4}=1$ then the area 1 square units of the triangle OPQ (where O is the origin) is equal to ___

84. Tangents are drawn from the point (h,k) to the hyperbola $3x^2-2y^2=6$ and inclined an angles α and β to the x-axis. If $(\tan\alpha)(\tan\beta)=2$ then the value of $2h^2-k^2$ is ____

85. Let $x^2+y^2=4r^2$ and xy=1 intersect at A and B in the first quadrant. If $AB=\sqrt{14}$ units then the value of $|r|$=___

86. For hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, distance between the foci is 10 units. From the point $\left(2,\sqrt{3}\right)$, perpendicular tangents are drawn to the hyperbola, then the value of $\left|\frac{b}{a}\right|$ is ____

87. If the points of intersection of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the circle $x^2+y^2=4b, b>4$ lie on the curve $y^2=3x^2$, then b is equal to ___

88. If the circle $x^2+y^2=a^2$ intersects the hyperbola $xy=c^2$ at four points $P(x_1,y_1)$, $Q(x_2,y_2)$, $R(x_3,y_3)$, $S(x_4,y_4)$ then $(x_1+x_2+x_3+x_4)+(y_1+y_2+y_3+y_4)-(x_1.x_2.x_3.x_4)+(y_1.y_2.y_3.y_4)=$ ___

89. P is any point on the hyperbola $x^2-y^2=a^2$ if F_1 and F_2 are the foci of the hyperbola and $(PF_1)(PF_2)=\lambda(OP)^2$, where O is the origin , then λ is equal to ___

90. If the locus of the foot of the perpendicular drawn from the centre upon any tangent to the ellipse $\frac{x^2}{40}+\frac{y^2}{10}=1$ is $\left(x^2+y^2\right)^2=px^2+qy^2$ then $\frac{p}{q}$ is equal to __